

Name: _____ SCH4U Unit 2 Organic Chemistry

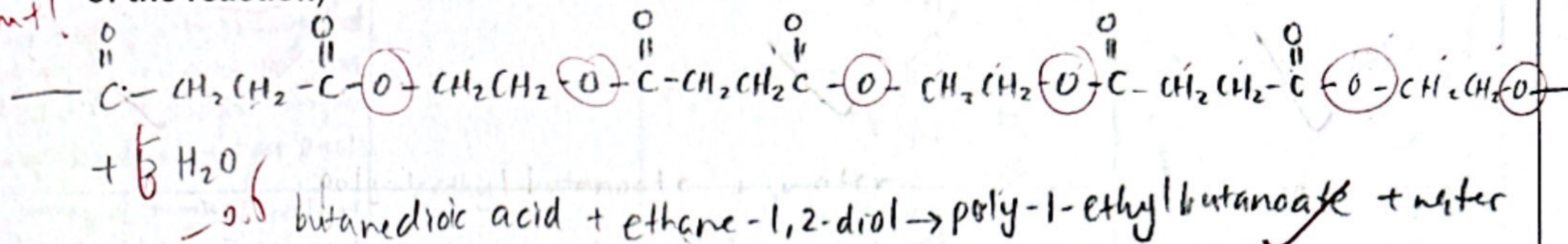
KNOWLEDGE /15	INQUIRY 13 /16	COMMUNICATION 10 /10	APPLICATION 9.5 /10
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PART A: MULTIPLE CHOICE KNOWLEDGE (15 MARKS)

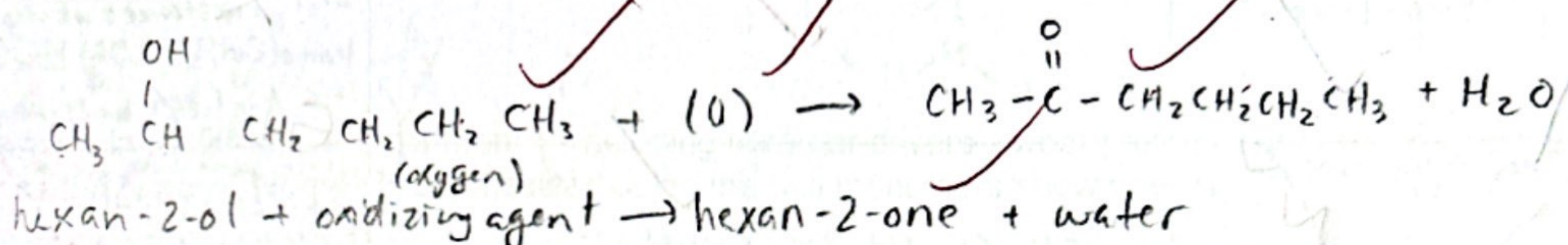
PART B: SHORT ANSWER QUESTIONS INQUIRY (16 MARKS)

1. Write a **chemical equation**, using structural formula or carbon skeleton, for each of the following reactions and **name the reactants and products**: (10 marks)

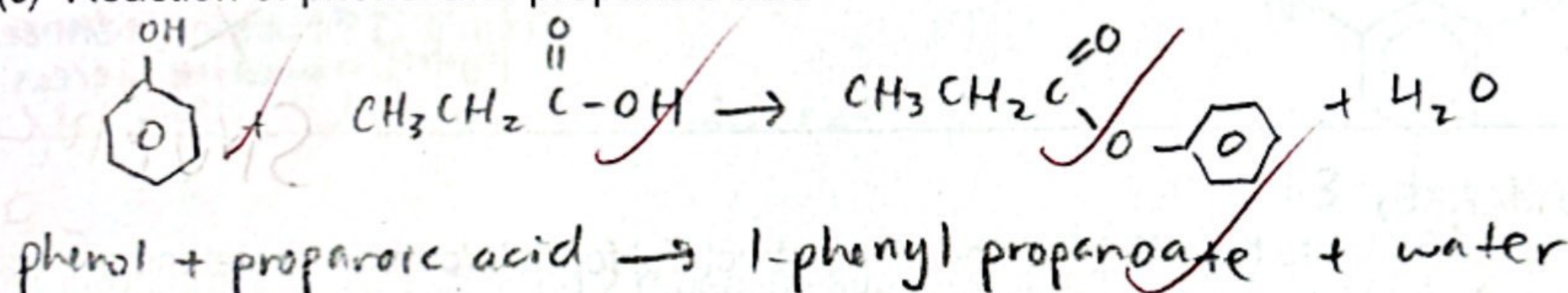
(a) Polymerization of butanedioic acid and ethane-1,2-diol. (show 3 units and just show the product(s) of the reaction)



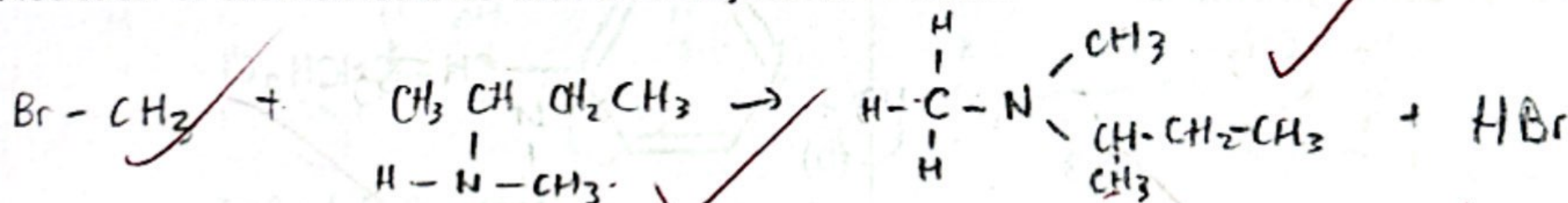
(b) Controlled oxidation of hexan-2-ol:



(c) Reaction of phenol and propanoic acid

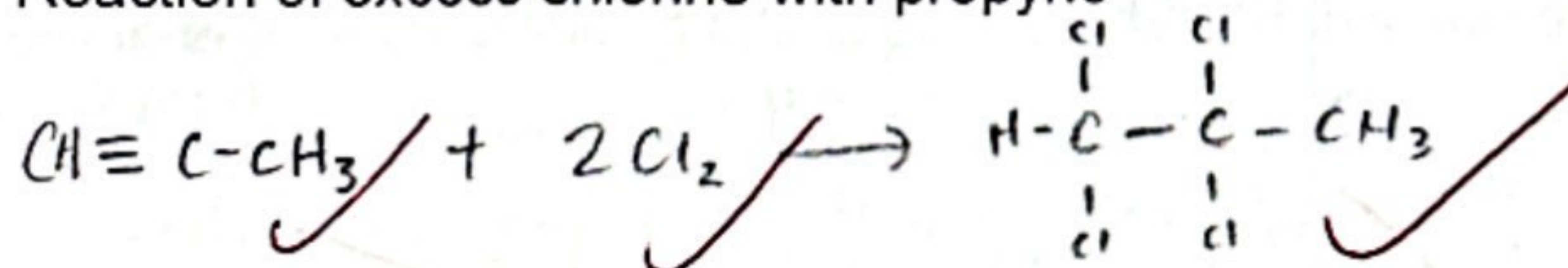


(d) Reaction of bromomethane with N-methylbutan-2-amine



bromo methane + N-methylbutan-2-amine $\rightarrow$ N,N-dimethyl butan-2-amine + hydrogen bromide

(e) Reaction of excess chlorine with propyne



propyne + excess chlorine $\rightarrow$ 1,1,2,2-tetrachloropropane

2. Use the following table to rank the compounds listed below in order of increasing boiling point

(i.e. (1) lowest boiling point → (4) highest boiling point) For each of the compounds:

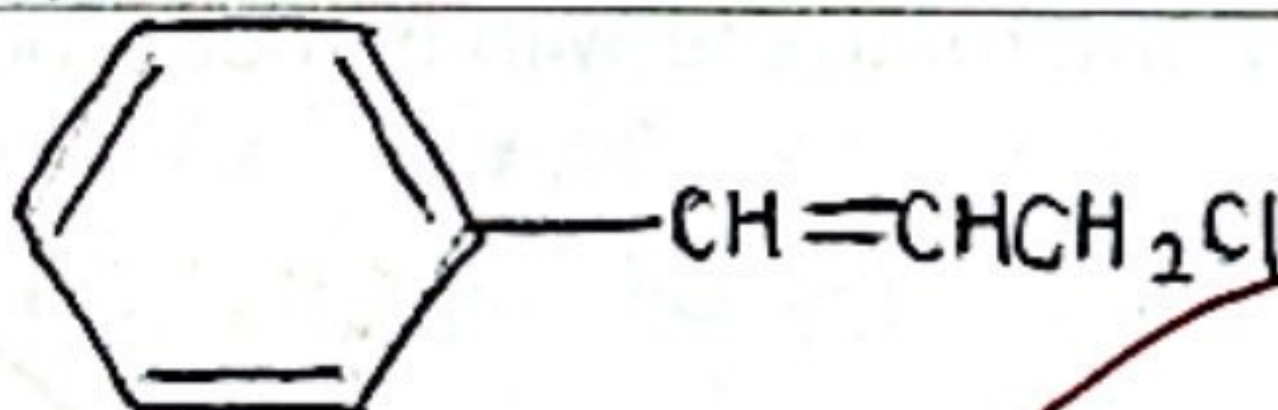
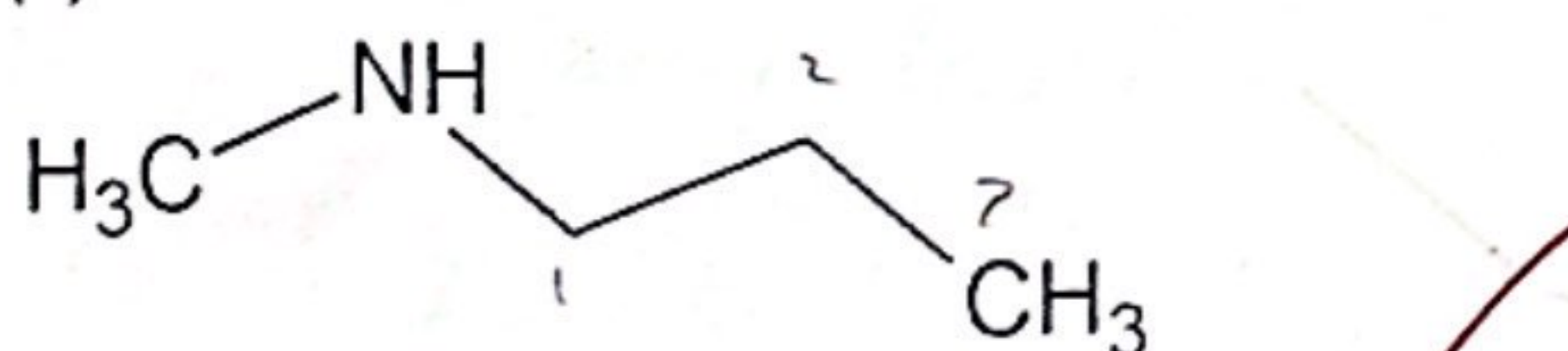
- Draw the structural formula
- Explain how this polarity determines its place on the list by referring to structure and IMF's (6 marks)

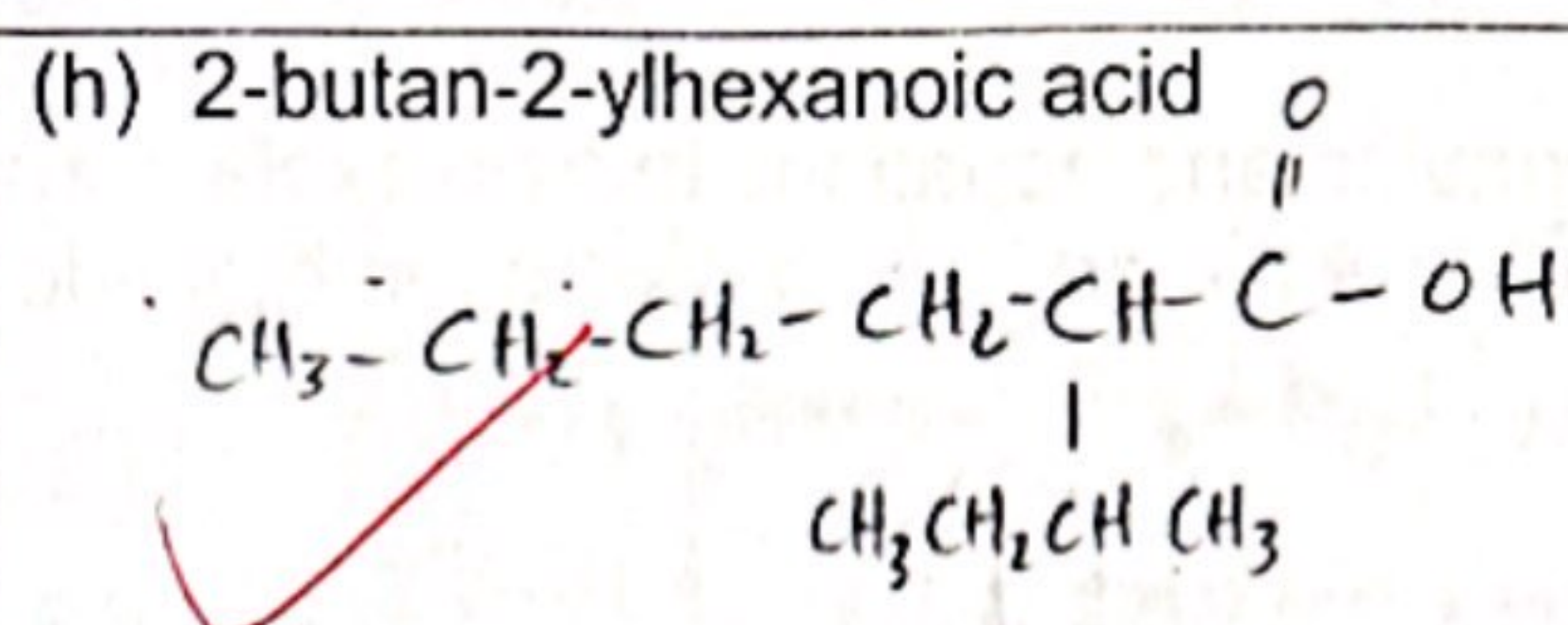
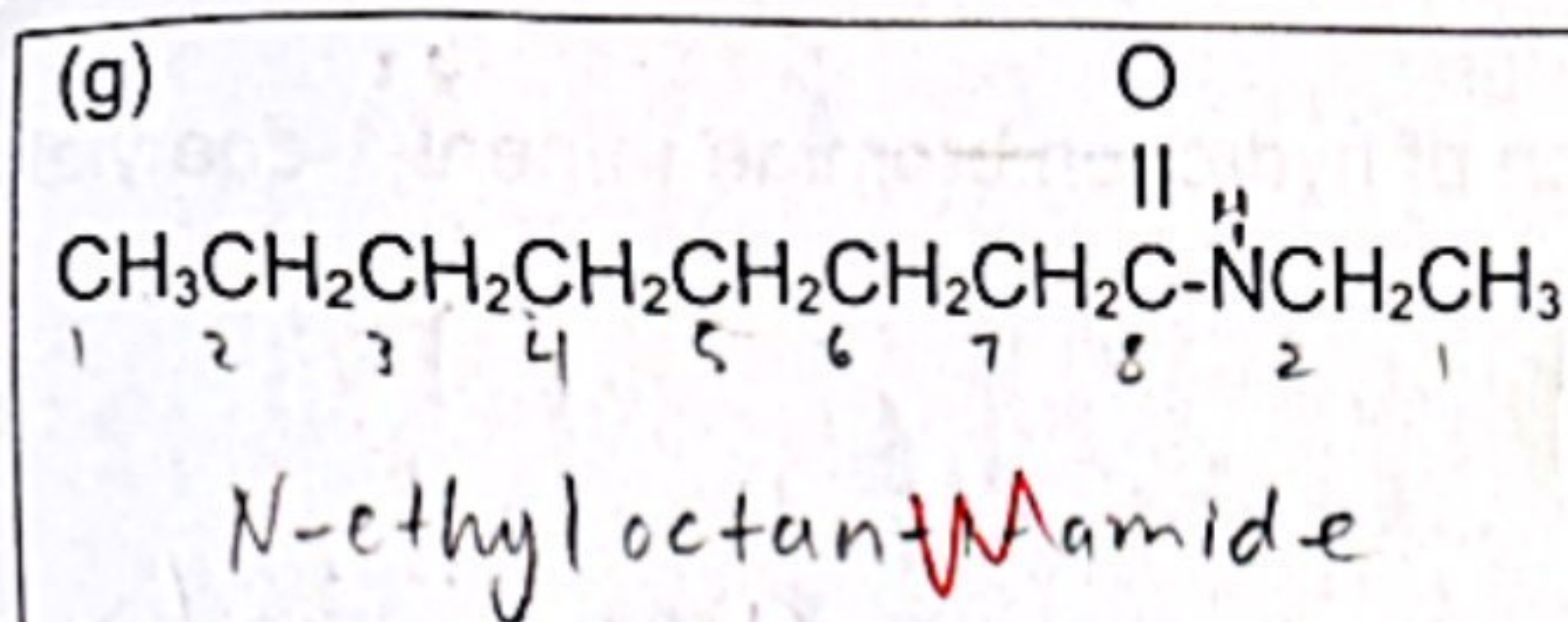
pentan-1-amine, pentanoic acid, pentan-2-one

NAME OF COMPOUND (LEAST TO MOST POLAR)	STRUCTURAL FORMULA/ CARBON SKELETON	EXPLANATION
(1) Lowest boiling point pentan-2-one	$\begin{array}{c} \text{O} \\ \\ \text{CH}_3 - \text{C} - \text{CH}_2 - \text{CH}_2 - \text{CH}_3 \end{array}$	the structure of pentan-2-one makes it a polar molecule but allowing for it to have dipole-dipole forces, LDF, but it does not have hydrogen bonding
(2) 3. pentan-1-amine	$\begin{array}{c} \text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3 \\ \\ \text{H} - \text{N} - \text{H} \end{array}$	Pentan-1-amine is polar and has LDF, dipole-dipole forces and 2 instances of hydrogen bonding, allowing it to be the highest melting point
(3) Highest boiling point 2. pentanoic acid	$\begin{array}{c} \text{O} \\ \\ \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{C} - \text{OH} \end{array}$	While pentanoic acid is polar and has LDF, dipole-dipole forces and hydrogen bonding, it only has one instance of hydrogen bonding whereas pentan-1-amine has 2

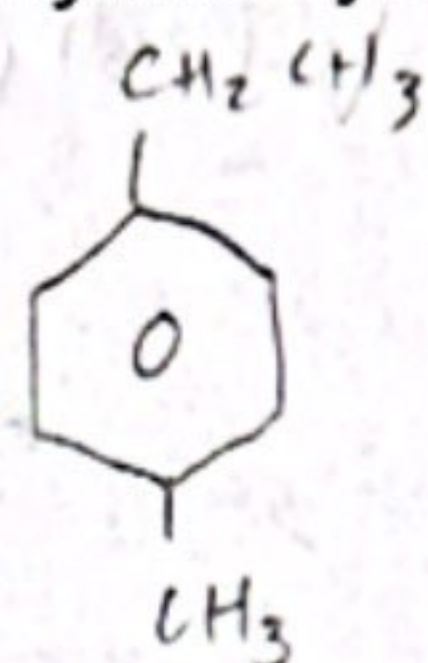
COMMUNICATION (10 MARKS)

1. Complete the following table: Write the IUPAC name of the structural formula/carbon skeletons OR draw the structural formula corresponding to each name. (10 marks)

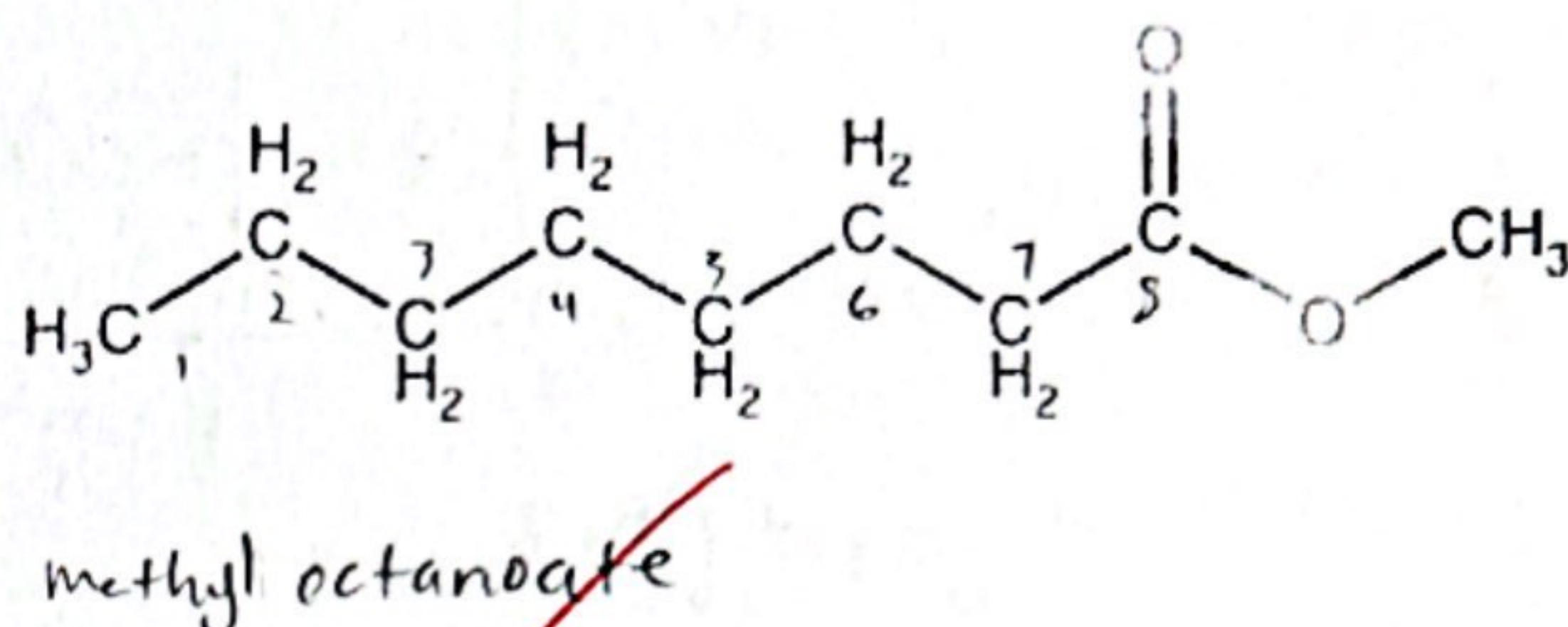
(a) 3-fluoro-3,4-dimethylpentanal $\begin{array}{c} \text{O} \\ \\ \text{CH} - \text{CH}_2 - \text{C} - \text{CH} - \text{CH}_3 \\ \quad \\ \text{F} \quad \text{CH}_3 \end{array}$	(b)  3-chloro-1-phenylprop-1-ene
(c)  N-methylpropan-1-amine	(d) a 2° alcohol with the formula C ₄ H ₁₀ O $\begin{array}{c} \text{OH} \\ \\ \text{CH}_3 - \text{CH} - \text{CH}_2 - \text{CH}_3 \end{array}$
(e) propan-2-yl heptanoate 2-propyl heptanoate $\begin{array}{c} \text{O} \\ \\ \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{C} - \text{O} - \text{CH}(\text{CH}_3)_2 \end{array}$	(f) $\begin{array}{c} \text{CH}_2\text{CH}_3 \quad \text{O} \\ \quad \\ \text{CH}_3 - \text{CH} - \text{CH}_3 - \text{C} - \text{CH}_3 \\ 4 \quad 3 \quad 2 \quad 1 \end{array}$ 4-methylhexan-2-one



(i) p-ethylmethylbenzene



(j)



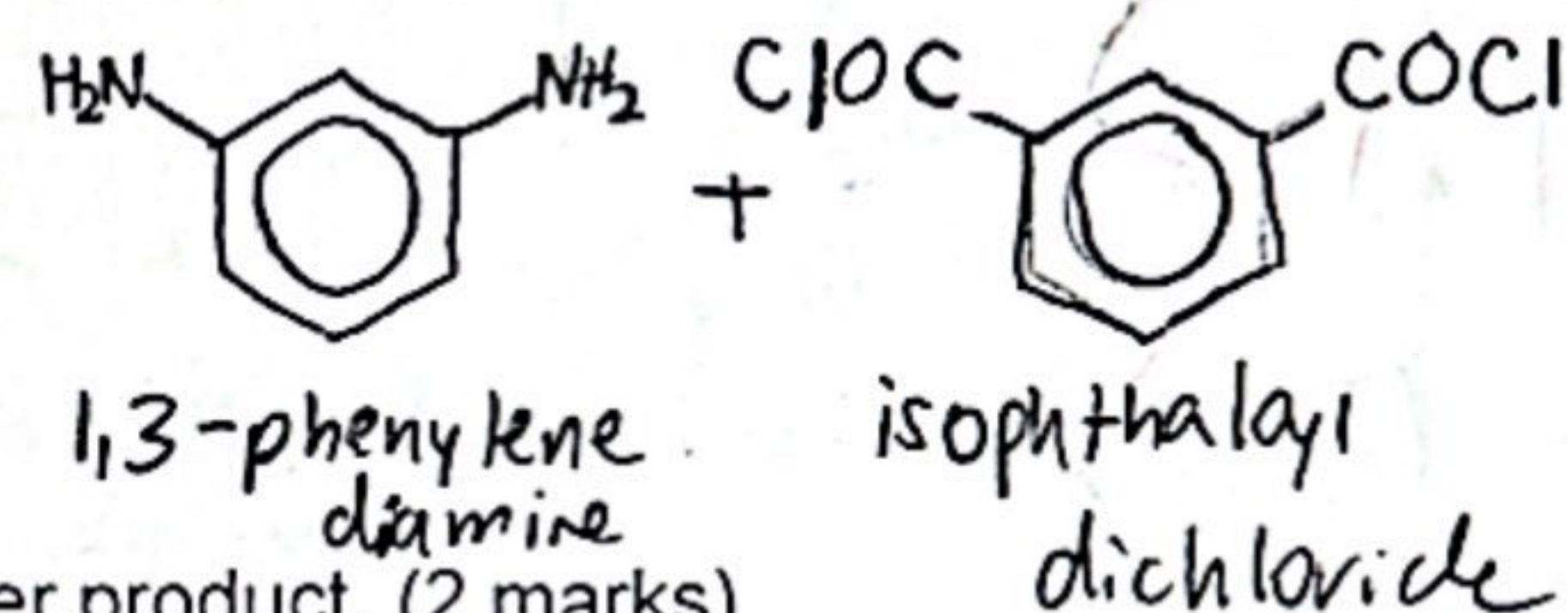
APPLICATION (10 MARKS)



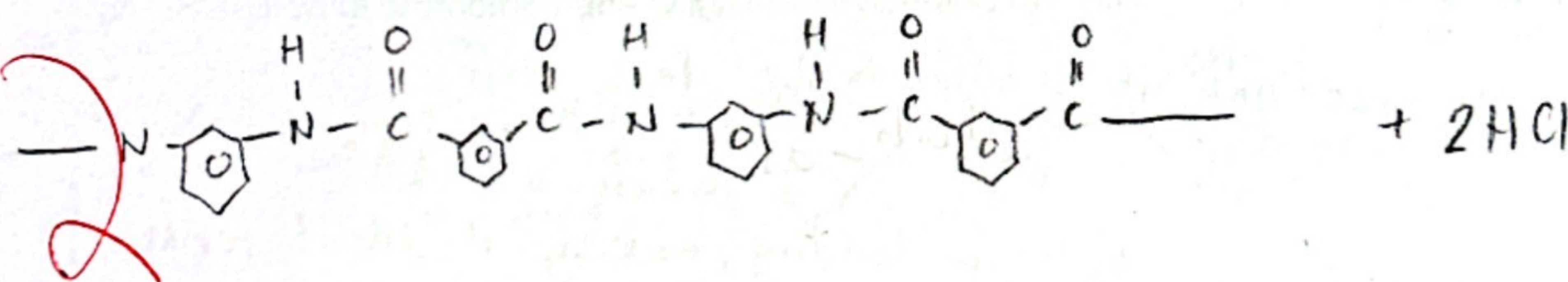
1. Many workers in hazardous environments—including race-car drivers—wear protective clothing made from a polymer called Nomex. Nomex is synthesized using the two monomers shown below. (5 marks total)

(a) Will there be a by-product of the polymerization reaction? If so, what is the by-product? (1 mark)

yes - hydrogen chloride is produced



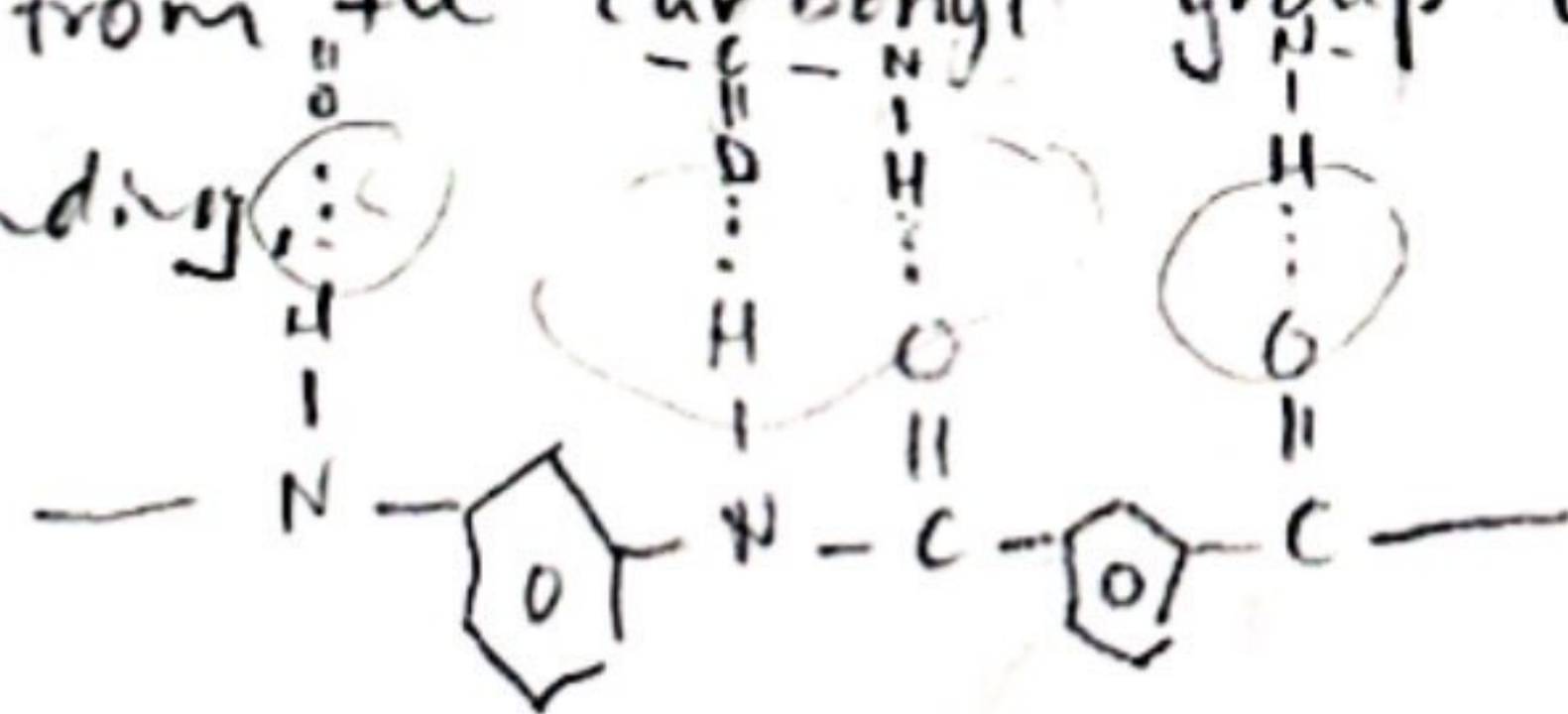
(b) Draw a structural diagram showing 2 repeating subunits of the polymer product. (2 marks)



(c) If you were to enable crosslinking in this polymer, would this improve the material? Explain how the crosslinking would occur. (2 marks)

Yes, when crosslinking happens, strong IMF forces (hydrogen bonding), hold the layers of polymers together, meaning that the clothing would be more durable and strong.

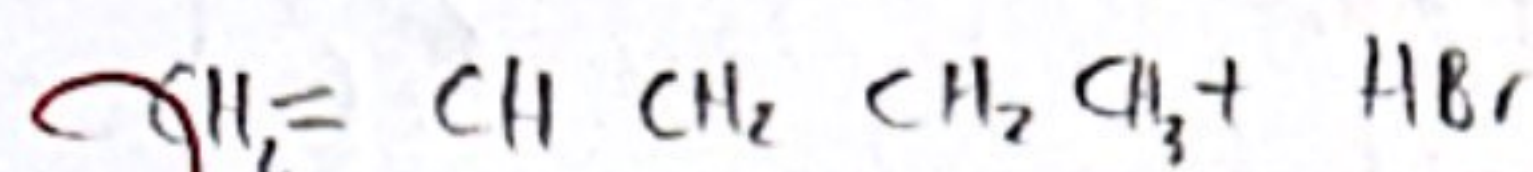
The crosslink would occur via the hydrogen attached to nitrogen bonding with the oxygen from the carbonyl group when the stack on top of each other via hydrogen bonding.



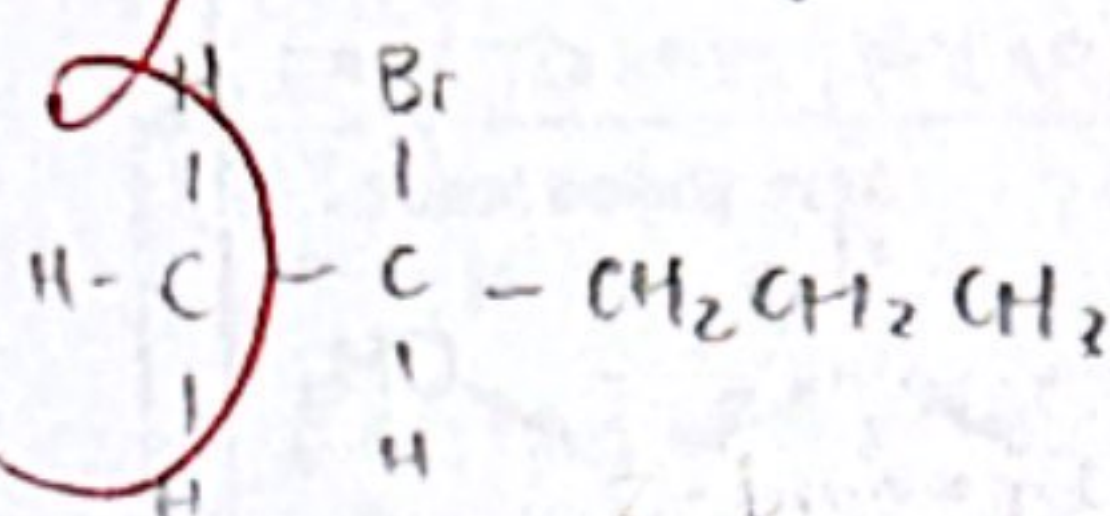
★ you can ignore markovnikov here only

2. Use structural formulas and equations to help explain why addition of hydrogen bromide to pent-1-ene yields two different products but the addition of hydrogen bromide to pent-2-ene yields only one. (3 marks)

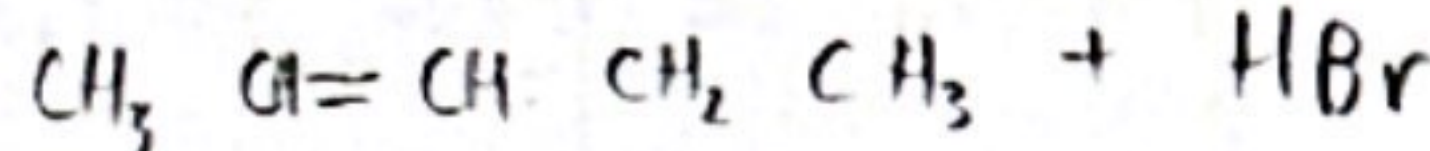
pent-1-ene + hydrogen bromide
→ 2-bromopentane



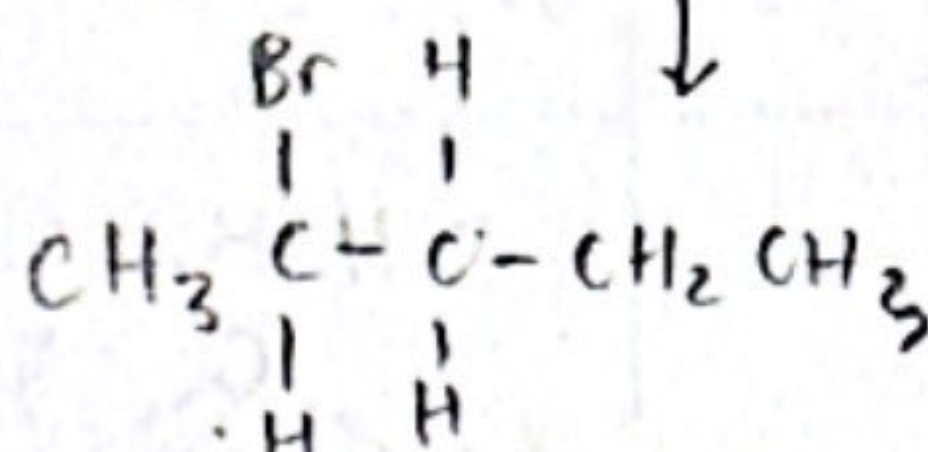
↓



pent-2-ene + hydrogen bromide
→ 2-bromopentane



↓



answered
vertically

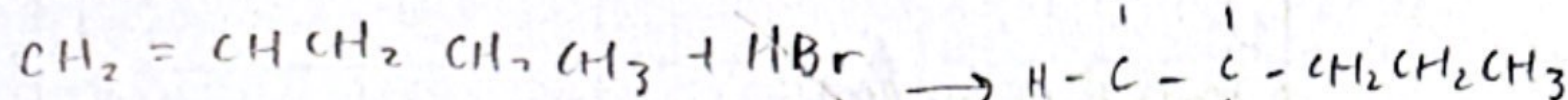
if it was but-2-ene there'd only be one product

since it is symmetrical

and you'd just get 2-bromobutane

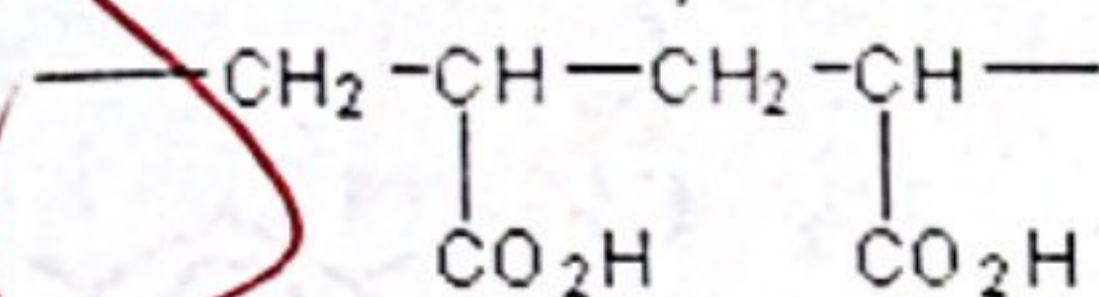
whereas but-1-ene can do

both 1-bromobutane and 2-bromobutane

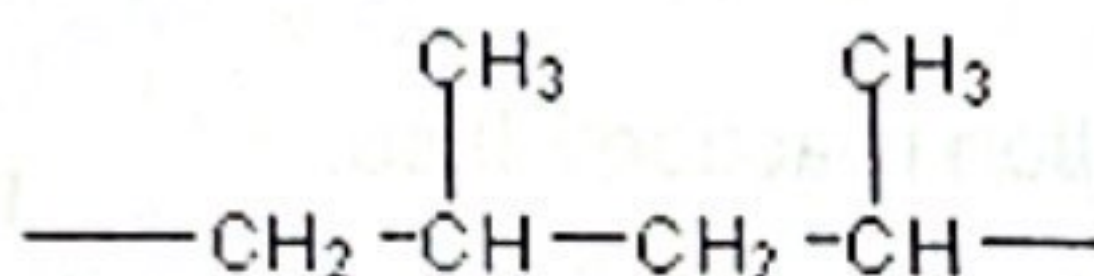


pent-1-ene + hydrogen bromide → 1-bromopentane

3. In the design of a new baby diaper, the manufacturer uses two polymers. The structure of these molecules is given below. Which type of polymer is best suited to the outside of the diaper and which to the inside? Explain your reasoning. (2 marks)



(I)



(II)

(I) is more suited for the outside and (II) is more suited for the inside

The outside of a diaper needs to be stretchy and pliable and since

(II) is unable to crosslink, it is going to be less stiff.

label - 0.5

(I) is able to crosslink via hydrogen bonding, making it ideal for the inside as you need to absorb water for the insides

It can take water in between the hydrogen bonding, allowing for water absorption

